Evaluation of Intra-entity Level Unlearning Quality

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***Abstract*—Large language models (LLMs) can memorize undesirable information during pre-training stage, motivating the development of machine unlearning methods that remove specific knowledge without retraining the model from scratch. Existing unlearning benchmarks mostly evaluate unlearning at the entity level: remove all facts about an entity. However, real-world scenarios often require more fine-grained forgetting. In this work, we study intra-entity unlearning, where the model must forget targeted fact within a multi-fact profile containing entangled knowledge. We construct a synthetic evaluation dataset with correlated facts and evaluate three common unlearning methods: gradient ascent, gradient difference, and negative preference optimization. Our experiments show that while these methods can suppress the target fact, they often significantly degrade performance on correlated retained facts more than facts with less correlation. Representation-level analysis further reveals that unlearning primarily modifies late-layer representations and diverges from a retain-only oracle model. These results highlight limitations of current unlearning approaches when applied to entangled knowledge.**

***Index Terms*—Machine Unlearning, Large Language Models, Representation Analysis**

I. Introduction

Large language models (LLMs) can unconsciously learn and memorize information that should not be retained during training, such as private or copyrighted content. Due to the impractical cost of retraining a model completely, machine unlearning, which aims to erase the model's memory of the sensitive knowledge while keeping the model functioning as before, has been considered as a feasible solution [3].

There are several machine unlearning methods proposed. Gradient ascent is a typical and intuitive method, where the model unlearns by maximizing the loss on the forget set [3]. As an improvement over gradient ascent, which often harms the model's original utility and results in poor forgetting performance, gradient difference optimizes both the forget and retain losses [5]. Other than gradient-based approaches, Negative Preference Optimization (NPO) removes undesired content by minimizing the probability of the model generating the forgetting materials, which is often more stable [2]. These are the three methods focused on in this paper for unlearning-related tasks.

Benchmarks are essential for evaluating LLM unlearning. The TOFU (Task of Fictitious Unlearning) benchmark is one evaluation benchmark that is widely recognized. The authors generated a dataset of 200 synthetic author profiles, each with 20 Q&A pairs. Then, the authors use metrics (ROUGE-L, Truth Ratio, Forget Quality) to capture model utility. In their experiments, the researchers evaluated unlearning methods such as GA and NPO, and concluded that existing methods cannot unlearn information without harming the model's overall utility [1]. MUSE (Machine Unlearning Six-Way Evaluation) realized that current unlearning benchmarks are not holistic enough and have narrow evaluation dimensions. Therefore, they proposed six key dimensions of machine unlearning and experimented with eight common unlearning methods on forgetting tasks to evaluate performance on their benchmark. They had similar comments on the unlearning methods they tested as those concluded by TOFU [6].

However, current benchmarks ignore the fact that real-world data are always entangled and related. When performing unlearning, the model tends to unlearn the entire entity and all the facts regarding it, thus facts that are supposed to be retained are also forgotten. This can make the unlearning outcomes look good and satisfying, but the evaluation result is actually misleading and unpersuasive [4]. Under the setting of TOFU, where some of the author facts are usually tied to other facts, this limitation becomes more significant. Thus, our goal is to first build an intra-entity unlearning dataset that incorporates more diverse bioprofiles and fact entanglement, then allow models to unlearn a specific fact that is entangled with another remaining fact and evaluate the unlearning outcomes.

II. Dataset Design and Generation

To study *intra-entity* unlearning, we construct a synthetic benchmark in which each person is associated with a rich, multi-fact profile rather than a single isolated statement. The dataset contains 20 fictitious public figures with diverse names, regions, and occupations. For each person, we generate 20 facts that together form a coherent profile. Among these facts, exactly one is designated as the forget target: the `negative_incident`. All remaining facts are treated as retained knowledge. This setup is meant to model a more realistic unlearning scenario, where the goal is not to erase an entire entity from the model, but to remove one specific memory while preserving the rest of what the model knows about that same person.

This design differs fundamentally from TOFU [1]. In TOFU [1], forgetting happens at the entity level: a subset of authors is selected, and all facts about those authors are removed from

TABLE I: Concrete example of fact generation and QA conversion for a single entity (Anya Petrova-Ríos). **Top:** The structured profile facts, demonstrating intentional entanglement (the retained breakthrough event references the "Terra Disputa" anchor). **Bottom:** The corresponding conversion into QA pairs, split into the retain set (SR) and the forget set (SF).

Part A: Structured Facts

Field	Anchor	Fact
breakthrough_year_event	Raízes Publication	Raízes Publication: Anya's impactful book, published by Editora Raízes, gained recognition following the complex 'Terra Disputa' land rights case.
negative_incident	Terra Disputa	Terra Disputa: Anya Petrova-Ríos faced public criticism after 'Casa da Floresta' funding was briefly suspended by the Bahia Cultural Council.
...	...	...

Part B: QA Pairs and Set Allocation

Field	Set	Q	A
breakthrough_year_event	retain	What does "Raízes Publication" refer to in Anya Petrova-Ríos's profile?	Anya's impactful book, published by Editora Raízes, gained recognition following the complex "Terra Disputa" land rights case.
negative_incident	forget	What does "Terra Disputa" refer to in Anya Petrova-Ríos's profile?	Anya Petrova-Ríos faced public criticism after "Casa da Floresta" funding was briefly suspended by the Bahia Cultural Council.
...	...	...	...

the model. In our evaluation, forgetting happens at the fact level within each entity. Every person appears in both the forget and retain evaluation setting: one fact is marked for forgetting, while the other facts about that same person should still be answerable. This change makes the task more fine-grained and substantially harder. A model can no longer solve the problem by broadly suppressing everything associated with one identity. Instead, it must separate one targeted fact from neighboring facts that belong to the same profile.

Another important feature of our dataset is that the facts are not fully independent. We intentionally introduce entangled fact structure so that the forget target is connected to nearby retained facts. For example, some retained fields are written to reference the same underlying event anchor as the negative_incident, while others are designed to be semantically close but not equivalent. This matters because it creates the exact tension we want to evaluate later: strong unlearning should suppress the target fact itself, but it should not unnecessarily damage related retained information. In other words, the benchmark is designed to expose over-forgetting as well as under-forgetting.

The dataset structure also aligns directly with our experimental protocol. From the 20 facts for each person, we create one QA pair per fact. The forget set, denoted by SF, contains exactly one QA pair per person, corresponding to the negative_incident. The retain set, denoted by SR, contains the other 19 QA pairs for that same person. This split gives a clean evaluation interface: forgetting performance is measured on SF, while utility is measured on SR. In addition, we generate paraphrased evaluation questions so that the model is tested on semantic generalization rather than only exact training wording. For forget-quality style evaluation, we also build a dedicated forget-set file that includes, for each forget example, one paraphrased correct answer and several closely matched wrong answers. This lets us evaluate whether the model truly loses preference for the forgotten fact, rather than merely memorizing one surface form.

The generation pipeline consists of four stages. Throughout the entire pipeline, we use Gemma-3-27B-IT as the generation model. In Stage 1, we generate synthetic profiles with diverse yet coherent identities. In Stage 2, we expand each profile into 20 structured and partially entangled facts. At this stage we also conduct red-team checking to identify leakage, ambiguity, contradictions, and overly easy shortcuts that could weaken the forget-versus-retain distinction. In Stage 3, we convert these facts into QA pairs and split them into SF and SR, while also generating paraphrased evaluation questions. In Stage 4, we build the forget-set evaluation examples used in truth-ratio and forget-quality style metrics. The final benchmark is therefore designed to match our experimental goal: evaluating whether an unlearning method can remove one targeted fact while preserving the rest of the entity's knowledge.

III. Experiment Setup

The base model used in our experiments for unlearning is Qwen2.5-1.5B-Instruct. We fine-tuned the base model using LoRA with rank 16. Through this process, all data in both the forget and retain sets are trained, so the model contains all the knowledge. Similarly, we fine-tuned the base model with only the retain set and obtained a retain-only model (OR). This model is seen as a reference for the ultimate unlearned model. Both models illustrate a Rouge-L score of 1, which suggests full absorption of the knowledge. Next, we ran GA, GD, and NPO with different epochs on the fine-tuned model with both the forget and retain sets to unlearn the knowledge

in the forget set. We use learning rate of 1e-5 for all methods and use $\beta = 0.1$ for NPO.

IV. Evaluation Framework

Our evaluation metrics are divided into two different categories: the regular output level metrics focus on the logits assigned to each token, similar to TOFU [1], and the inner model representation level metric.

A. Output Level Metrics

ROUGE-L: It measures the level of longest common subsequence matching between two texts. Higher Rouge-L means better matching. We evaluated the Rouge-L score on both the forget set and the retain set.

Truth Ratio: The ratio of the length-scaled probability of a correct answer to the mean of five wrong answers. Wrong answers share a similar structure to the correct answer. A truth ratio much smaller than 1 indicates that the model strongly prefers the correct answer, and a truth ratio greater than one means the model favors the wrong answers, truth ratio near one means the model is uncertain.

Model Utility: harmonic mean of ROUGE-L and probability scores on the held-out real-author set (RA), and a world-facts set (WF).

B. Representation-Level Metrics

For a model M and a prompt q, we apply the chat template and compute the mean-pooled hidden state over all prompt tokens at a chosen layer: $\mathbf{h} = \frac{\sum_{t \in T}, \mathbf{H}^{(l)} t}{|T|}$

where $\mathbf{H}^{(l)}$ denotes the l-th transformer layer. Let $\mathbf{h}_{\mathrm{UL}}$, $\mathbf{h}_{\mathrm{FT}}$, $\mathbf{h}_{\mathrm{OR}}$ be the resulting vectors for a given prompt. We define:

- $\cos_{\mathrm{FT}}$: cosine similarity between $\mathbf{h}_{\mathrm{UL}}$ and $\mathbf{h}_{\mathrm{FT}}$. This measures the representation level similarity between the unlearned model and fine tuned base model that contains all knowledge.
- $\cos_{OR}$: cosine similarity between $\mathbf{h}_{\mathrm{UL}}$ and $\mathbf{h}_{\mathrm{OR}}$. This measures the representation level similarity between the unlearned model and the model fine-tuned on only the retain set. The retain model is the best 'unlearned' model since it never saw the data before. A good unlearned model should share a similar representation as the retain only model, so it has a high cos_{OR}.
- $\Delta h = |\mathbf{h}_{\mathrm{UL}} - \mathbf{h}_{\mathrm{FT}}|_2$: hidden-state drift magnitude.

V. Experiment Results

A. Forgetting–Retention Trade-off

Table II establishes baselines: the retain-only oracle (OR) and fine-tuned model (FT), in addition with all three methods at epoch 5. Table III shows the dynamics of method in 5–20 unlearning epochs.

It is worth noting that the truth ratio on the retain-only model is slightly below one, which indicates that it slightly prefers the correct answer. This is due to our dataset design: In the retain set, there are some facts we intentionally add strong correlation to the negative forget facts. Therefore, it is possible that the model saw similar words during fine tuning on the retain set and assign high probability to it when see it again on the forget set.

TABLE II: Baseline and ep5 results. SF↓ = forget ROUGE-L. SR↑ = retain ROUGE-L. Util = model utility. TR<1.0 means the model still prefers the correct answer.

Model	SF R↓	SF TR↑	SR R↑	Util↑
Retain-only (OR)	0.128	0.749	1.000	0.474
Fine-tuned (FT)	1.000	0.119	1.000	0.456
GA (ep5)	0.329	0.188	0.994	0.434
GD (ep5)	0.301	0.138	0.995	0.398
NPO (ep5)	0.288	0.170	0.981	0.428

TABLE III: Epoch progression: SF ROUGE-L (lower↓), SF Truth Ratio (higher↑; TR$\gg$1 indicates strong output-level forgetting), SR ROUGE-L (higher↑), and model utility (higher↑).

Method	Ep	SF R↓	SF TR↑	SR R↑	Util↑
GA	5	0.329	0.19	0.994	0.434
	10	0.192	13.7	0.621	0.220
	15	0.160	92.8	0.298	0.157
	20	0.079	2.1×10^5	0.099	0.074
GD	5	0.301	0.14	0.995	0.398
	10	0.219	2.07	0.827	0.342
	15	0.187	201	0.533	0.335
	20	0.149	2.7×10^3	0.577	0.304
NPO	5	0.288	0.17	0.981	0.428
	10	0.237	15.7	0.518	0.285
	15	0.129	63.2	0.262	0.145
	20	0.053	1.8×10^3	0.136	$<$0.001

At epoch5, all three methods reduce SF ROUGE-L from 1.000 to 0.288–0.329 while preserving SR above 0.981, with NPO achieving the best forgetting and GD the best retain quality on the retain set. The oracle's SF ROUGE=0.128 sets the gold-standard target; all methods remain substantially above it at ep5. In addition, the truth ratio is near 0.2 for all methods, which indicates that the model still strongly prefer the correct answer. The forget quality is very low.

Beyond epoch 5, the forget set truth ratio starts to surge. The model starts to assign a tiny probability to the correct answer compared to the perturbed answers, which indicates effective forgetting. However, for gradient ascent and negative preference optimization, the retain quality and model utility suffer a lot. The strong correlation between forget facts and part of retain facts lead to a huge decrease in retain quality. For comparison, the gradient difference method is more stable due to the effective calibration of the retain loss.

B. Representation Analysis: Last Layer

Table IV reports representation metrics averaged across all forget-set questions. The baseline $\cos(\mathrm{FT}, \mathrm{OR})$ is 0.960, providing context for the oracle-proximity scores.

We expect good unlearning will lead to a decrease in $\cos_{FT}$ and an increase in $\cos_{OR}$. However, for all methods $\cos_{OR}$ and

TABLE IV: Last-layer representation similarity metrics. $\cos(\mathrm{UL}, \mathrm{FT})$: proximity to fine-tuned model. $\cos(\mathrm{UL}, \mathrm{OR})$: proximity to oracle. Baseline: $\cos(\mathrm{FT}, \mathrm{OR})$=0.960.

Method	Ep	SF R	$\cos_{\mathrm{FT}}$	$\cos_{\mathrm{OR}}$
GA	5	0.329	0.993	0.949
	10	0.192	0.970	0.922
	15	0.160	0.947	0.898
	20	0.079	0.868	0.814
GD	5	0.301	0.995	0.951
	10	0.219	0.980	0.933
	15	0.187	0.953	0.904
	20	0.149	0.940	0.889
NPO	5	0.288	0.993	0.949
	10	0.237	0.962	0.912
	15	0.129	0.937	0.887
	20	0.053	0.835	0.783

$\cos_{FT}$ decrease monotonically with increase in the number of epochs.

Since the cosine similarity between the retain-only model and fine tuned baseline is 0.960, heavier unlearning moves representations further from the retain-only model. In addition, the consistent decrease means that more training epochs don't magically lead to a better unlearned model.

C. Representation Drift Across Layers

In addition to the final representation, we also want to know how the unlearning algorithm modifies each layer. Therefore, we measure the representation drift $\Delta h = \|\mathbf{h}_{\mathrm{UL}} - \mathbf{h}_{\mathrm{FT}}\|_2$ between the unlearned model and the fine-tuned base model. Figure 1 shows the results.

It is not surprising that more epochs of unlearning lead to larger drift in hidden representation. More interestingly, the representation drift increases a lot in late layers like layer 26-28, while representation drifts are relatively small and stable in early layers. This suggests that the effects of these three are concentrated primarily in late-layer representations, with early layers largely preserved.

In addition, the largest drift occurs at layer 27 instead of the final layer. The final layer appears to reduce the representation difference between the unlearned model and the fine-tuned base model. One possible explanation is that the final transformer layer acts as a projection toward the unembedding layer and output token space, which may smooth or regularize intermediate perturbations introduced by unlearning.

D. Correlated Retain Degradation

One goal of our intra-entity level dataset design is to manually introduce strong knowledge entanglement and test how knowledge entanglement increase the difficulty of retain knowledge preserving. Therefore, we test the retain quality for two groups of facts in our retain set. One group is called the correlated retain group since it has a strong correlation to the forget set. The correlated facts contain noun phrases, such as event names that co-occur with the forget set content.

TABLE V: Correlated vs. other retain-fact ROUGE-L. Gap = Corr − Other (negative means correlated are more damaged).

Model	SF R	Corr R↑	Other R↑	Gap
Retain-only	0.128	1.000	1.000	0.000
Fine-tuned	1.000	1.000	1.000	0.000
GA ep10	0.192	0.394	0.634	−0.240
GD ep10	0.219	0.443	0.849	−0.406
NPO ep10	0.237	0.239	0.534	−0.296
GA ep15	0.160	0.112	0.308	−0.196
GD ep15	0.187	0.218	0.551	−0.332
NPO ep15	0.129	0.081	0.272	−0.191
GA ep20	0.079	0.046	0.102	−0.056
GD ep20	0.149	0.265	0.595	−0.330
NPO ep20	0.053	0.042	0.141	−0.099

Table V describes our result. For these unlearned model with satisfactory forget quality (ROUGE-L match retain-only model and truth ratio greater than 1), the correlated facts degrading more than other facts.

GD consistently shows the largest gap despite having the best absolute retain ROUGE. This means that for the correlated facts, even GD can't find a balancing point. NPO at epoch 15 achieves SF ROUGE=0.129—matching the oracle's 0.128—while Reducing correlated retain to only 0.081 ROUGE. The gap narrows at epoch 20, but only because both categories collapse simultaneously as the model becomes less useful over correlated ones.

This experiment confirms our hypothesis that unlearning with higher knowledge entanglement is a more difficult task than unlearning with totally independent facts. We think that these three unlearning methods on the forget set increase the loss for some shared tokens, and inadvertently damage retain-fact generation paths that potentially share the same weights.

VI. Discussion

A. Inner representation cosine similarity as a diagnostic

The retain-only model provides a useful reference for the representation of a model that never observed the forgotten information, so we assume closer representation of unlearned model lead to better unlearning. However, all methods diverge from the oracle as forgetting deepens. This can explain the inevitable tradeoff between forget quality and retain quality/model utility we observed in the unlearning experiment. The decrease in $\cos_{\mathrm{FT}}$ suggests that unlearning make model's internal representation different from the fine tuned base model and lead to the forgetting behavior. However, the simultaneous decreases in cos_{OR} indicate that the unlearning algorithm failed to identify a direction in representation space that selectively removes the forgotten knowledge while preserving the remaining knowledge.

In addition, we observed that an increase in the number of epochs doesn't improve alignment with the oracle model. Instead, additional training only shifts the balance between forgetting strength and general model utility. Therefore, in

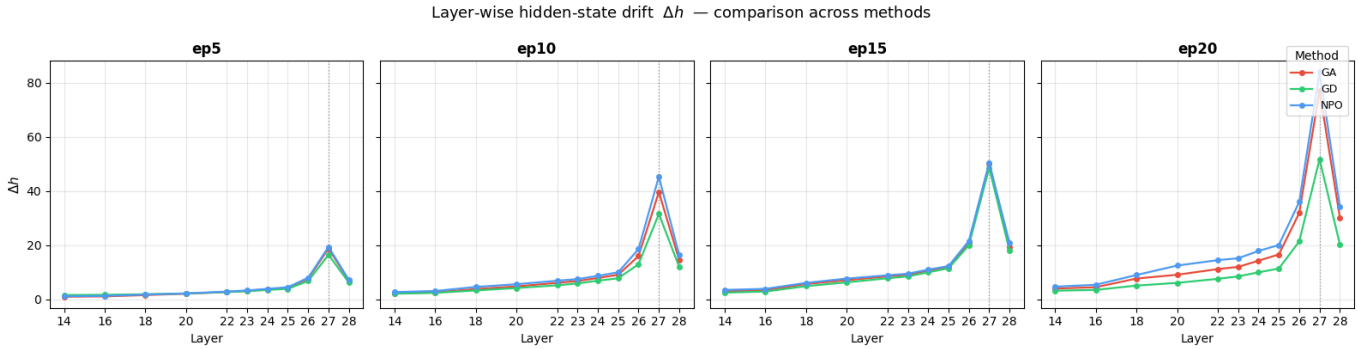

Fig. 1: Layer-wise hidden representation drift

a controlled unlearning experiment setting where the retain-only model is available, representation-level cosine similarity provides an efficient diagnostic metric. To be specific, it allows researchers to assess whether an unlearning method is moving the model toward the desired representation space using only a small number of training epochs, making it an efficient early indicator of unlearning quality.

B. Layer-wise representation drift

Our layer-wise analysis reveals that representation drift induced by unlearning is highly uneven across the different transformer blocks. Specifically, the magnitude of drift between the unlearned model and fined tuned base model remains small and stable in middle layers, while it increases significantly in the final third layers of the model.

Prior LLM interpretability work indicate a strong possibility that factual associations in transformer language models tend to be localized in middle feed-forward layers, which act as key–value memories for entity and relational knowledge [7], while late layers are more task specific. [9] In addition, other studies further demonstrate that modifying activations in certain middle-to-late layers can directly alter a model's stored factual associations without retraining the entire network [8].

Taken together with these findings, our layer-wise results suggest that the LoRA-based unlearning methods studied here mainly change late-stage representations that influence output decisions, instead of strongly altering the middle layers commonly associated with factual knowledge storage. This observation is consistent with the hypothesis that unlearning acts more like behavioral suppression than as true erasure of stored knowledge.

C. Comparison with existing benchmark and dataset sensitivity

Our strategy of using fine-tuning to absorb the knowledge we want to forget is similar to the TOFU benchmark [1], However, the mechanism and the measurable failure mode are different. In the TOFU benchmark task setting, unlearning deletes all facts about an entity, so there is no within-entity retain set. The correlated-retain degradation we report more than unrelated ones (Table V) is invisible to full-entity benchmarks. Studying the unlearning methods' capability at the intra-entity level is important since it defines a more realistic task. For example, some public figures want to remove defamatory information but preserve their general information from the model. In addition, we provide model representation level analysis, which is absent in existing benchmarks. Representation-level analysis provides a finer-grained view of how unlearning algorithms modify internal model states.

The knowledge entanglement we introduced to our dataset make unlearning more challenging task because retain quality is expected to be lower. Furthermore, we don't claim that we have found the best hyperparameters for the unlearning algorithm we tested. Therefore, we don't make a claim about which unlearning method is better. Instead, we want to show common failure modes of all three methods and provide analysis and explanations.

VII. Limitation

Due to time and computation limit, all of our experiments are using Lora fine tuning on relatively small Qwen2.5-1.5b-instruct model, so results may not transfer to other models. Our further work may explore different models with full weight fine tuning to test whether our results can transfer to more general cases.

In addition, the scale of our dataset is relatively small compared to what would be required for a full benchmark. Due to time and computational constraints, we constructed a limited number of synthetic entities and facts. Therefore, we view this work as an evaluation study rather than a finalized benchmark. However, we believe that increasing the size of the data set would make the unlearning task more challenging. As a result, the failure modes we observe in our small scale setting are unlikely to disappear in larger datasets.

VIII. Conclusion

In this project, we create a specially designed dataset to study machine unlearning under a more realistic setting where factual knowledge about an entity is partially entangled. Instead of removing all information about an entity, our task focuses on evaluating unlearning algorithms' ability to forget specific negative facts while preserving other related knowledge.

Our experiments reveal several consistent behaviors across the three unlearning methods we tested. First, although the algorithms are able to reduce the model's accuracy on the forget set, they also degrade performance on correlated retain facts more than on unrelated retain facts. Second, representation-level analysis shows that as forgetting progresses, the internal representations of the unlearned models diverge from the oracle retain-only model. Finally, our layer-wise representation drift analysis suggests that unlearning mainly modifies the model's late-layer representations while keeping earlier and middle layers relatively stable.

In conclusion, our results highlight several limitations of current unlearning approaches when applied to entangled knowledge. We hope that the dataset design and representation-level analysis presented in this work can help motivate future research on more precise and interpretable unlearning mechanisms.

Member Contributions

a) Woyu Li: Conducted unlearning experiments, generated dataset, and authored evaluation, results, and discussion sections.

b) Jiawei Zhang: Designed dataset logic and fact entanglement, wrote scripts of data generation pipeline, generated dataset and evaluation data, and authored dataset section.

c) Ernest Gao: Helped with running experiments to generate data, searching for and reviewing relevant research papers, and preparing the proposal and final report write-ups.

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